3. Optimizing Data Retrieval Techniques

1.

V1 → Advanced Rag and Retrieval Optimisation →

→ The rag paradigm is continaudy evolving,

→ A recent study survey categorised Rag into three stages:

1. Naive Rag
2. Advanced Rag
3. Modular Rag

→ Difference b/w Naive Rag and Advanced Rag →

→ Naive Rag → Naive rag research paradigm represents the earliest methodology which gain prominance shortly aftr the widespread of Chatgpt

- Naive Rag includes indexing, retrieval, augmentation and generation, these are the traditional process, and are also characterised as a retrieve read framework.

→ Advanced Rag → Advanced Rag was introduced specifically to overcome the limitation of rag

- Advanced Rag is actually a Naive Rag with enhanced retrieval quality and post-retrieval strategies.

→ Re-ranking system → ordering/sorting and ranking the document in the decreasing order of relevance.

Scanned with OKEN Scanner

→ Why Rag optimization is imp? →

Problems (without Optimisation	→ Optimization fixes
① AI might retrieve irrelevant or outdated info	→ Improves search quality and ranking
② Answers can be inaccurate or off topic	→ Ensure context-aware, precise responses
③ Retrieval can be slow	→ Makes it faster and cost-efficient
④ Large data can overwhelm the model	→ Add filters and embeddings to focus only on useful pieces

(V2) → Retrieval system optimisation →

→ Retrieval stages aim to identify the most relevant context.

→ Usually the retrieval is based on either vector search or hybrid search

Vector search	hybrid search
① Finds results based on meaning that is semantic similarities.	① Combines meaning that is semantic meaning + exact keywords matching
② Converts text into embeddings (vector embedding) and finds the nearest vector using cosine similarity or dot product.	② Runs both vector searches and keyword searches and merges the results using weighted scoring.

Scanned with OKEN Scanner

③ Used in (Best for) → NLP, semantic similarities general knowledge retrieval	③ Used in (Best for) → Real-world application needing both semantic relevance and exact matches (eg- enterprise, search, technical docs) ⑧
④ Accuracy is high for semantic meaning	④ Accuracy is ~~low~~ very high as it balances semantic and keyword precision.
⑤ faster, depends on vector size and index type	⑤ Slightly slower - perform two searches and merges result.
⑥ Simple to Implement	⑥ More complex
⑦ Ex → Faiss, Pinecone, Chroma, Milvus.	⑦ Weaviate, Elastic search hybrid mode, etc.
⑧ Summary → understands what you mean	⑧ Summary → understands what you means and exactly what you said.

→ Various Techniques for Optimising Retrieval system →

① Vector search →
→ Embedding model especially fine-tuned embedding models for domain-specific contexts
→ Vector search uses Dynamic embeddings for context in which words are used.

② Hybrid search → that combines vector search with keyword based search.

Scanned with OKEN Scanner

V3 → Setting up Environment

→ Step 1 → Installing Python Packages (Necessary)

→ Llama-index is used.

→ Llama-index is a simple, flexible, framework for connecting custom data sources to large language model.

→ Llama-index is open-source

→ Step 2 → Definning llm model, llm and data

V6 → Relevance Scoring and Re-ranking

→ re-ranking system → It is final step of optimisation.

→ This system helps in recalculating scores and sort the documents in the decreasing order of relevance to increase information pertinance for the llm.

→ Techniques used for relevance scoring and re-ranking method:→

① Basic Re-ranking and Filtering

② Advanced Re-ranking with Machine Learning

① Basic Re-ranking and Filtering →

→ Basic re-ranking sorts the documents by their already available course generated from the relevant ~~score~~ retrieval ~~step~~ system from the previous step.

→ filtering functions is applied to the sorting document to exclude any documents with relevant score below threshold value.

Scanned with OKEN Scanner

② Advanced Re-ranking with machine Learning

→ Advanced re-ranking with ml is more of a sophisticated approach where machine learning model are used for re-ranking documents. ⑨

→ The ml models are usually trained on the previous interactions b/w user and system allows ml to have an intelligent re-ranking where for a given user ml ranks the document taking into account the user's preference behaviours and contextual information.

Scanned with OKEN Scanner